

Linear Programming Course Summary

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1 Key Terms

- **Slack Variable:** A variable added to a less-than-or-equal-to inequality constraint to transform it into an equality constraint. The slack variable represents the difference between the left-hand side and the right-hand side of the inequality.
- **Feasible Region:** The set of all points that satisfy all the constraints of a linear programming problem.
- **Vertex:** A point where two or more constraint boundaries intersect in the feasible region of a linear programming problem. In the Simplex method, optimal solutions are found at vertices.
- **Basic Variable:** In the Simplex method, a variable that is expressed in terms of non-basic variables in a dictionary. Basic variables are set to values that satisfy the constraints.
- **Non-basic Variable:** In the Simplex method, a variable that is set to zero in a given iteration of the algorithm. The values of the basic variables are determined by the values of the non-basic variables.
- **Pivot Operation:** The process of exchanging a basic variable with a non-basic variable in the Simplex method, leading to a new dictionary and a movement to a different vertex in the feasible region.
- **Dictionary:** A representation of a linear programming problem in a tabular format, where basic variables are expressed as functions of non-basic variables. It is used in the Simplex method to systematically move from one vertex to another.
- **Auxiliary Problem:** A modified linear programming problem introduced to find an initial feasible solution when the origin is not feasible in the original problem. It involves adding a new variable to shift the feasible region.

2 Problem Types

- **Finding an Initial Feasible Solution:** The Simplex method requires an initial feasible solution. If the origin is not feasible, an auxiliary problem is introduced to find a feasible solution to the original problem.
- **Handling Infeasible Origins:** The origin $(0, 0)$ may not satisfy all constraints in a linear programming problem. Techniques are needed to find a feasible starting point for the Simplex method.
- **Optimizing with Slack Variables:** The Simplex method uses slack variables to convert inequality constraints into equalities, facilitating the iterative optimization process.
- **Graphical Representation of Feasible Regions:** Visualizing the feasible region (the set of points satisfying all constraints) helps understand the Simplex method's iterative movement between vertices.
- **Pivot Operations in the Simplex Method:** The Simplex method iteratively improves the solution by exchanging basic and non-basic variables (pivot operations) in the dictionary.
- **Working with Dictionaries of Variables:** The Simplex method uses a dictionary to represent the linear programming problem, where basic variables are expressed in terms of non-basic variables. Each dictionary corresponds to a vertex of the feasible region.
- **Solving Linear Programs with Three Variables:** Extending the Simplex method to problems with more than two variables requires careful consideration of the constraints and iterative optimization.

3 Algorithm Solutions

- **Simplex Method:** Iteratively move from one vertex of the feasible region to another by setting non-basic variables to zero and choosing a new set of (non)basic variables to improve the objective function value. This is equivalent to moving from vertex to vertex in the feasible region.
- **Auxiliary Problem:** When the origin is not feasible, introduce a new variable x_0 and modify the constraints to create an auxiliary problem that maximizes $-x_0$. Solve this auxiliary problem to find a feasible solution for the original problem. If the optimal solution has $x_0 = 0$, then a feasible solution to the original problem is found.